VES-9 Pressure Vessel Operating Manual (DOC-VM-094)

Section 1

1 Operating limits

Maximum allowable working pressure: 12.5 bar at 40 degC.
Between 40 and 150 degC the MAWP derates linearly to 9.8 bar at 150 degC.
Minimum design metal temperature: -29 degC (carbon steel, labelled flange).
Contents must remain non-corrosive; chloride limit 25 ppm.

Section 2

2 Safety valve

The primary safety valve is set at 13.8 bar and must never be re-set in the field. Check the set pressure every 12 months with the certified test bench TB-3. The rupture disc on the second nozzle bursts at 15.2 bar and must be replaced after any activation.

Section 3

3 Inspection intervals

Visual internal inspection every 24 months; crack test (dye penetrant) every 48 months around nozzles N1, N2 and N4. Wall thickness reading near the saddle is recorded each internal inspection; a reading below 8.6 mm requires engineering review before the next fill.

Section 4

4 Filling procedure

Vent the vessel, verify the safety valve isolation is fully open, fill to the sight glass 80 percent mark, then pressurize in 2 bar steps holding 5 minutes per step up to operating pressure. Retain the pressure log 5 years.